Visual Kubernetes Documentation Guide

Beginner → Advanced • Linux + Windows • Labs + Troubleshooting

ShadowX Lab — Enterprise DevOps Academy

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1. Introduction to Kubernetes

Kubernetes (K8s) is an **orchestrator**: you describe the desired state in YAML, and K8s makes the cluster match it — scheduling containers, restarting failures, scaling replicas, and load-balancing traffic.

Containers vs Virtual Machines

	Virtual Machine	Container
Boot time	minutes	seconds
Isolation	full OS	process + namespaces
Size	GBs	MBs
Density	tens / host	hundreds / host

Docker vs Kubernetes

Docker **builds and runs** containers on one host. Kubernetes **orchestrates** containers across many hosts.

Lab 1.1 — install Docker and run the first container

```
# Ubuntu / Debian
sudo apt-get update && sudo apt-get install -y docker.io
sudo systemctl enable --now docker
sudo docker run --rm hello-world
```

Expected output:

```
Hello from Docker!
This message shows that your installation appears to be working correctly.
```

[NOTE] If you see 'permission denied on /var/run/docker.sock' — add your user to the docker group: sudo usermod -aG docker $USER then log out and back in.

2. Linux basics for Kubernetes

Before installing Kubernetes, every Linux node needs a clean baseline.

Hostname	Unique per node, resolvable. Used as the node name.
Static IP	Required — pod networking breaks if the node IP changes.
DNS	Must resolve the API server and external registries.
Swap	Must be disabled — kubelet refuses to start with swap on.
Firewall	Open 6443 (API), 10250 (kubelet), 2379-80 (etcd), 30000-32767 (NodePort).
systemctl	Used to enable/start kubelet and containerd.

Lab 2.1 — prepare a Linux node

```
sudo hostnamectl set-hostname k8s-cp-01
sudo swapoff -a
sudo sed -i '/ swap / s/^/#/' /etc/fstab
sudo modprobe br_netfilter
echo 'br_netfilter' | sudo tee /etc/modules-load.d/k8s.conf
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.ipv4.ip_forward = 1
EOF
sudo sysctl --system
```

Troubleshooting

Symptom	Likely cause	Fix
kubelet won't start	swap still on	swapoff -a + edit fstab
nodes can't pull image	DNS broken	check /etc/resolv.conf, nslookup
pods can't reach service	firewalld blocking	open NodePort + 10250
nodes can't join	API port closed	open TCP 6443 on control plane

3. Windows basics for Kubernetes

Windows nodes can run **process-isolated** or **Hyper-V isolated** containers, but they must join a Linux control plane.

Windows Server	2019 LTSC or 2022 — same build as the container image.
Hyper-V	Optional, enables stronger isolation per pod.
PowerShell	Primary admin tool — replaces bash for setup.
HNS	Host Network Service — Windows analogue of Linux CNI.
containerd	Recommended runtime since Kubernetes 1.24.

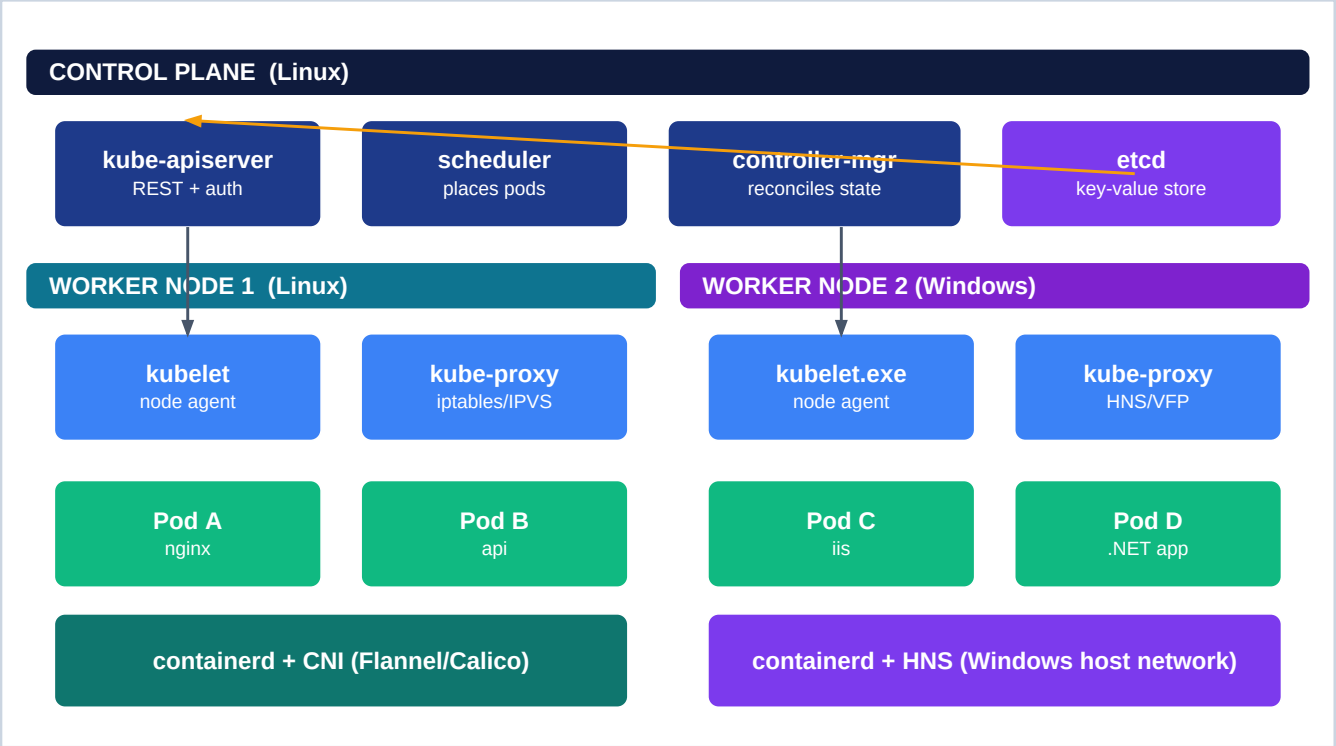
Lab 3.1 — enable containers on Windows Server

```
# PowerShell (as Administrator)
Install-WindowsFeature -Name Containers
Restart-Computer -Force
# After reboot:
curl.exe -LO https://github.com/containerd/containerd/releases/download/v1.7.20/containerd-1.
7.20-windows-amd64.tar.gz
tar xzf containerd-1.7.20-windows-amd64.tar.gz -C $env:ProgramFiles\containerd
& "$env:ProgramFiles\containerd\containerd.exe" --register-service
Start-Service containerd
```

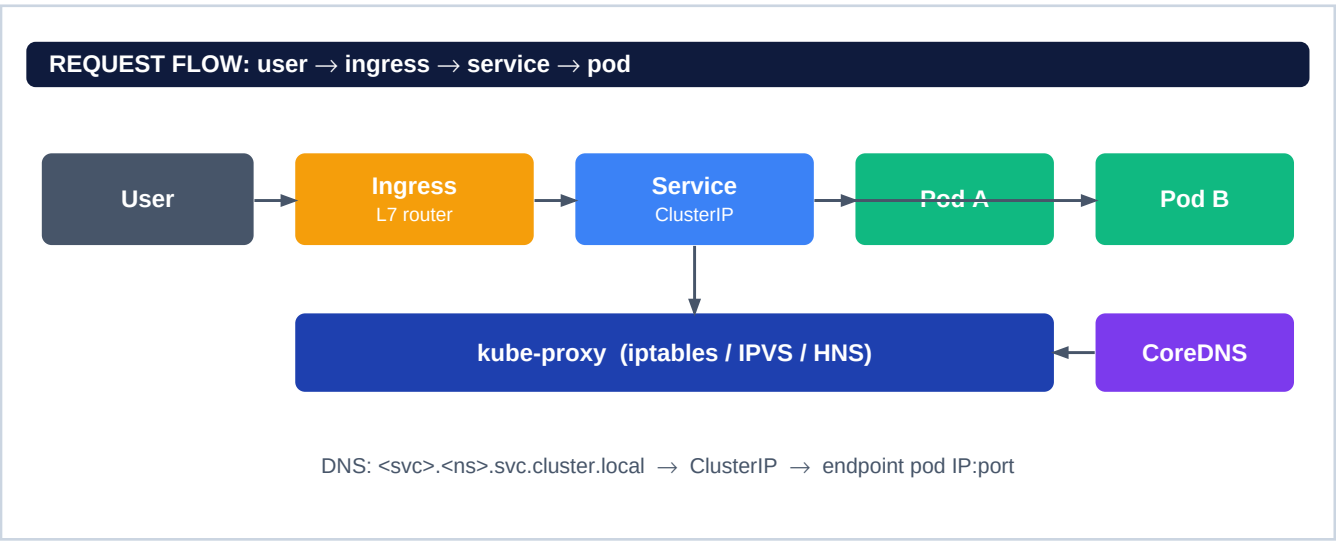
[NOTE] Windows containers must match the host build (1809 / ltsc2022). Mismatched base images fail with *'container OS version mismatch'*.

4. Kubernetes architecture

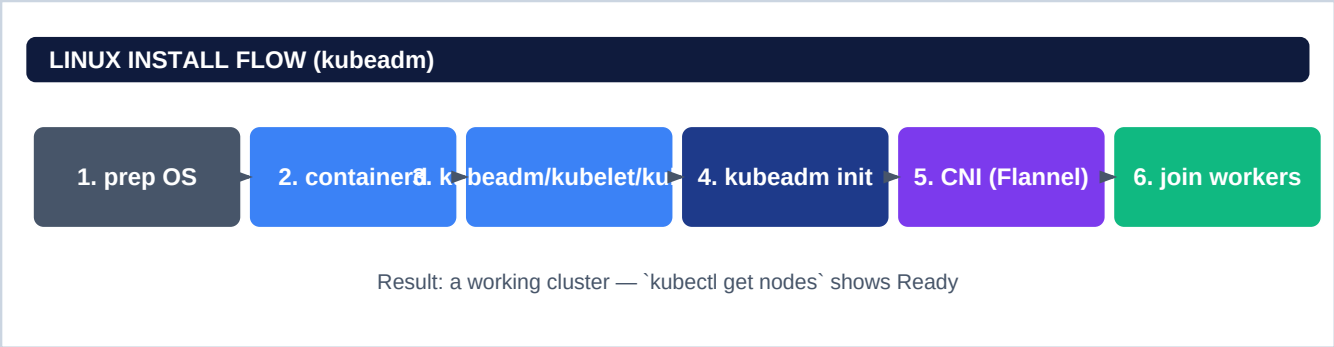
A cluster has a **control plane** (the brain) and one or more **worker nodes** (the muscle). The control plane only schedules — your pods actually run on workers.



kube-apiserver	Front door. Every kubectl / controller talks REST here.
scheduler	Picks which node a new pod runs on.
controller-mgr	Reconciles desired vs actual state (ReplicaSet, Node, etc.).
etcd	Distributed key-value store — the single source of truth.
kubelet	Per-node agent. Talks to the runtime, runs pods, reports status.
kube-proxy	Implements Services via iptables / IPVS / HNS rules.
CoreDNS	Cluster DNS — resolves Service names to ClusterIPs.



5. Install Kubernetes on Linux (kubeadm)



Lab 5.1 — install the runtime

```
sudo apt-get update && sudo apt-get install -y containerd
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml
sudo sed -i 's/SystemdCgroup = false/SystemdCgroup = true/' /etc/containerd/config.toml
sudo systemctl restart containerd
```

Lab 5.2 — install kubeadm, kubelet, kubectl

```
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | \
sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg]
https://pkgs.k8s.io/core:/stable:/v1.30/deb/ /' | \
sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo apt-get update && sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
```

Lab 5.3 — initialize the control plane and install Flannel

```
sudo kubeadm init --pod-network-cidr=10.244.0.0/16
mkdir -p $HOME/.kube && sudo cp /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
kubectl apply -f
https://raw.githubusercontent.com/flannel-io/flannel/master/Documentation/kube-flannel.yml
```

[TIP] Save the kubeadm join … line that kubeadm init prints — you'll need it for every worker. Regenerate later with kubeadm token create --print-join-command.

Troubleshooting kubeadm init

Symptom	Cause	Fix
[ERROR Swap]	swap still on	swapoff -a; comment in fstab
pulling images failed	no internet / no registry mirror	set HTTP_PROXY or use a local mirror
kubelet not running	cgroup driver mismatch	set SystemdCgroup=true in containerd
CNI not ready	Flannel/Calico not applied	kubectl apply -f <cni-manifest>

6. Build worker nodes

Linux worker — Lab 6.1

```
# On the worker, after installing containerd + kubeadm + kubelet:
sudo kubeadm join 10.0.0.10:6443 --token <token> \
--discovery-token-ca-cert-hash sha256:<hash>
# Verify from control plane:
kubectl get nodes -o wide
```

Windows worker — Lab 6.2

```
# PowerShell on the Windows node
curl.exe -LO
https://github.com/kubernetes-sigs/sig-windows-tools/releases/latest/download/PrepareNode.ps1
.\PrepareNode.ps1 -KubernetesVersion v1.30.0
kubeadm join 10.0.0.10:6443 --token <token> `
--discovery-token-ca-cert-hash sha256:<hash>
```

LINUX NODE		WINDOWS NODE	
Runtime:	containerd + runc	**Runtime:**	containerd + hcsshim
Network:	CNI (Flannel/Calico)	**Network:**	HNS + VFP
Agent:	kubelet (systemd)	**Agent:**	kubelet.exe (service)
Isolation:	namespaces + cgroups	**Isolation:**	Job objects / Hyper-V
Image OS:	Linux base (alpine)	**Image OS:**	windows/servercore/nano

Hybrid rule: control plane must run on Linux; Windows joins as workers only

[NOTE] The control plane **must** run on Linux. Windows nodes only join as workers, and you must add a nodeSelector (kubernetes.io/os: windows) to schedule Windows pods.

7. kubectl command reference

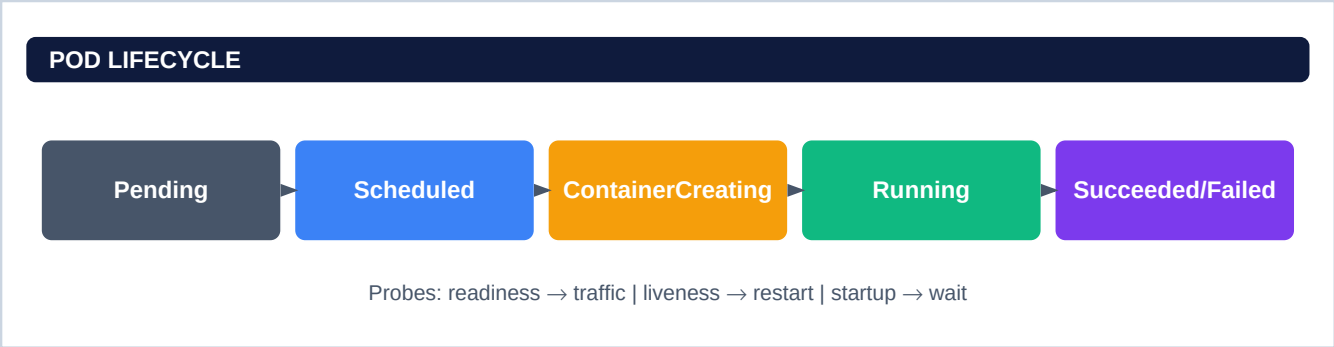
Command	Purpose	Example
get	list resources	kubectl get pods -A
describe	deep details + events	kubectl describe pod web-0
logs	container stdout/stderr	kubectl logs web-0 -f
exec	shell into a container	kubectl exec -it web-0 -- sh
apply	declarative create/update	kubectl apply -f deploy.yaml
delete	remove a resource	kubectl delete pod web-0
scale	change replicas	kubectl scale deploy web --replicas=5
rollout	update / rollback	kubectl rollout undo deploy/web
top	live CPU / memory	kubectl top pod

Reading kubectl get pods

```
NAME READY STATUS RESTARTS AGE
web-7d4-abc 1/1 Running 0 5m
api-86f-xyz 0/1 Pending 0 2m
```

READY = containers ready / total. STATUS = lifecycle phase. RESTARTS hints at crash loops. AGE = time since creation.

8. Pods and Deployments



Lab 8.1 — first Deployment

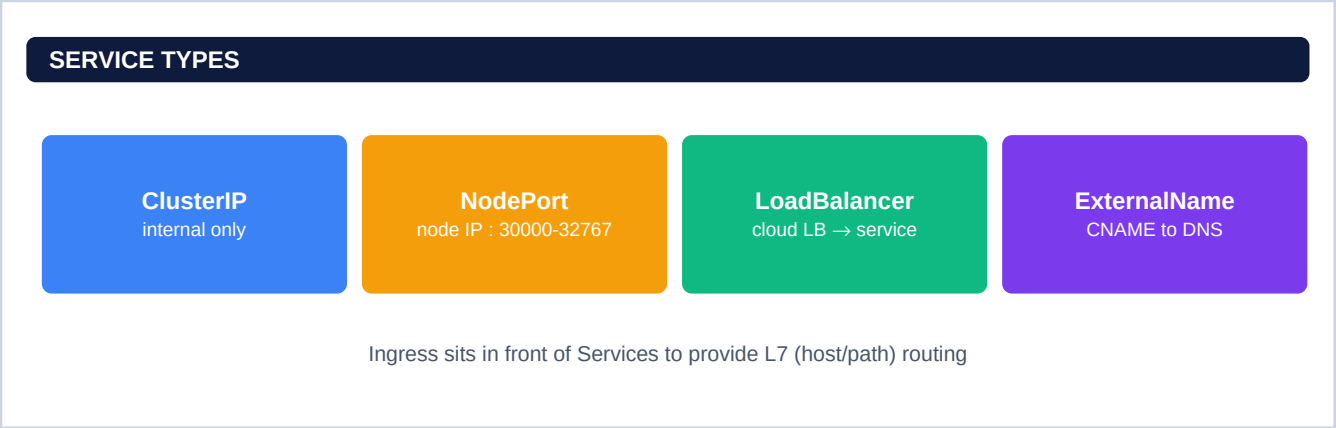
```
apiVersion: apps/v1
kind: Deployment
metadata: { name: web }
spec:
replicas: 3
selector: { matchLabels: { app: web } }
template:
metadata: { labels: { app: web } }
spec:
containers:
- name: nginx
image: nginx:1.27
ports: [{ containerPort: 80 }]
readinessProbe:
httpGet: { path: /, port: 80 }
```

```
kubectl apply -f web.yaml
kubectl get pods -l app=web
kubectl scale deploy web --replicas=5
kubectl set image deploy/web nginx=nginx:1.27.2 # rolling update
```

Troubleshooting pod states

State	Meaning	First check
Pending	no node fits	kubectl describe pod / kubectl get events
ImagePullBackOff	image name or auth wrong	check image tag + registry secret
CrashLoopBackOff	container exits repeatedly	kubectl logs --previous
OOMKilled	exceeded memory limit	raise resources.limits.memory
Error	command failed	kubectl logs + describe

9. Networking and Services



Lab 9.1 — expose the deployment

```
kubectl expose deploy web --port=80 --target-port=80 --type=ClusterIP
kubectl run busy --rm -it --image=busybox -- wget -qO- web
```

Lab 9.2 — NodePort for quick external access

```
kubectl expose deploy web --port=80 --type=NodePort
kubectl get svc web # note the 3xxxx port
curl http://<node-ip>:<nodeport>
```

DNS resolution inside the cluster

```
# A pod resolves another service via:
<service>.<namespace>.svc.cluster.local
```

[TIP] Use kubectl exec -it <pod> -- nslookup web to validate CoreDNS from inside the cluster.

10. Storage — Volumes, PV, PVC

Volume	Lives with the pod; gone when the pod dies.
PersistentVolume (PV)	Cluster-wide piece of storage provisioned by admin or driver.
PersistentVolumeClaim	A pod's request for storage of size / class.
StorageClass	How dynamic PVs are created (e.g. AWS gp3, Azure managed).

Lab 10.1 — dynamic persistent storage

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata: { name: data }
spec:
accessModes: [ReadWriteOnce]
resources: { requests: { storage: 5Gi } }
storageClassName: standard
```

```
kubectl apply -f pvc.yaml
kubectl get pvc
```

11. Security basics

RBAC	Role + RoleBinding grant verbs on resources to users / SAs.
ServiceAccount	Identity used by pods to talk to the API.
Secret	Base64-encoded sensitive data (mount or env).
NetworkPolicy	Pod-level firewall (requires a CNI that enforces them).

Lab 11.1 — read-only role

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata: { namespace: default, name: pod-reader }
rules:
- apiGroups: [""]
resources: [pods]
verbs: [get, list, watch]
```

[PITFALL] Secrets are **not encrypted by default** — enable encryption at rest in the API server, or use an external KMS / external-secrets operator.

12. Monitoring and observability

Prometheus	Scrapes /metrics endpoints across the cluster, stores time-series.
Grafana	Dashboards on top of Prometheus.
metrics-server	Powers kubectl top — short-lived in-memory metrics.
Loki / EFK	Centralised pod logs.

Quick install (kube-prometheus-stack)

```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm install kps prometheus-community/kube-prometheus-stack -n monitoring --create-namespace
kubectl -n monitoring port-forward svc/kps-grafana 3000:80
```

13. Troubleshooting playbook

Use this loop for almost every issue:

```
kubectl get … # what's the state?
kubectl describe … # why? read Events at the bottom
kubectl logs … --previous# what did it say before crashing?
kubectl get events -A # cluster-wide signal
kubectl exec -it … -- sh # poke from inside
```

Symptom	Likely cause	Commands / fix
Node NotReady	kubelet stopped, CNI missing	systemctl status kubelet; journalctl -u kubelet; kubectl describe node
Pod Pending	no schedulable node / PVC unbound	kubectl describe pod (Events); check taints, resources, storageClass
CrashLoopBackOff	app exits with non-zero	kubectl logs --previous; fix command/env
ImagePullBackOff	bad tag or no auth	kubectl describe pod; create docker-registry Secret
DNS failure	CoreDNS down or NetworkPolicy	kubectl -n kube-system get pods; nslookup from a debug pod
API server down	etcd unhealthy / cert expired	sudo crictl ps | grep apiserver; check /var/log/pods; kubeadm certs renew all
CNI issue	plugin pods crashing	kubectl -n kube-system get pods; reapply CNI manifest

14. Enterprise examples

Banking	Strict RBAC, NetworkPolicy default-deny, encrypted etcd, audit logs to SIEM.
SOC	Falco for runtime detection, audit log → Splunk, image signing with cosign.
Healthcare	HIPAA: encrypted PVs, isolated namespaces per tenant, pod security 'restricted'.
Hybrid	Linux control plane on-prem, Windows workers on Azure, WireGuard between sites.

Pattern — multi-tier app on Kubernetes

```
Ingress → Service(api) → Deployment(api, 3 replicas)
→ Service(db) → StatefulSet(postgres) + PVC
```

15. Cheat sheets

kubectl

```
kubectl get pods -A -o wide
kubectl describe node <name>
kubectl logs <pod> -c <container> -f --previous
kubectl exec -it <pod> -- bash
kubectl apply -f . / -k overlays/prod
kubectl rollout status deploy/web
kubectl rollout undo deploy/web
kubectl scale deploy web --replicas=10
kubectl port-forward svc/web 8080:80
kubectl top pod / kubectl top node
kubectl get events --sort-by=.lastTimestamp
```

kubeadm

```
kubeadm init --pod-network-cidr=10.244.0.0/16
kubeadm token create --print-join-command
kubeadm certs check-expiration
kubeadm certs renew all
kubeadm reset -f
```

Troubleshooting one-liners

```
kubectl get pods -A --field-selector=status.phase!=Running
kubectl get events -A --sort-by=.lastTimestamp | tail -20
kubectl -n kube-system logs -l k8s-app=kube-dns
sudo journalctl -u kubelet -e --no-pager
sudo crictl ps -a | grep -v Running
```

End of guide — happy shipping.